# Knowledge organiser · extraction steps

Today I can

Explain release, filtering and precipitation steps.

Record an observation without claiming purity.

State one practical limit or useful comparison.

Extraction means separating material from cells. A membrane is a boundary around a cell or an organelle.

Mashing breaks up soft fruit tissue. Detergent helps disrupt cell and nuclear membranes. This releases a mixture of cell contents.

Salt assists extraction and precipitation. Filtering removes larger pieces of tissue. The liquid that passes through is the filtrate.

Cold ethanol helps DNA precipitate. A precipitate is material that comes out of a solution. The trained adult handles and layers the ethanol.

Use only the approved method. No tasting or direct pupil contact with chemicals. A supplied-observation route is an equal choice.

Keep each role clear. Say what each step does. Use one step at a time.

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Version _____ 1

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# Knowledge organiser · observation and limits

Today I can

Explain release, filtering and precipitation steps.

Record an observation without claiming purity.

State one practical limit or useful comparison.

An observation records what was seen. Say where the material was and what it looked like.

White strands may be consistent with a DNA-containing precipitate. This uses method knowledge as well as sight. It is an interpretation.

A crude extract is a mixture that has not been purified. Visible material may include other substances. Appearance alone does not establish purity.

A water comparison helps test whether strands appeared in every tube. It does not identify every substance in the fruit sample.

Do not use visible amount as a true measure of DNA yield. Keep fruit mass, volumes, mixing and timing fair. An appropriate DNA measurement is needed for true yield.

Fruit-cell chromosomes contain DNA and genes. This extraction cannot show an individual gene, base sequence or double helix to the naked eye.

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